

J1

5V INPUT



D1

1N5819



C1

100nF 50V



#FLG01

PWR_FLAG



#FLG02

PWR_FLAG

AND: series NAND plus inverter and LED driver



SW1

B3F-4050



R1

10k 5%



R3

10k 5%



Q1

2N3904



R13

2.2k 5%



R5

10k 5%



Q3

2N3904



R14

2.2k 5%



R6

10k 5%



Q4

2N3904



R18

1.5k 5%



D2

TLHR5200



SW2

B3F-4050



R2

10k 5%



R4

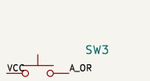
10k 5%



Q2

2N3904

OR: parallel NOR plus inverter and LED driver



SW3

B3F-4050



R21

10k 5%



R7

10k 5%



Q5

2N3904



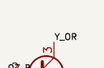
R15

2.2k 5%



R9

10k 5%



Q7

2N3904



R16

2.2k 5%



R10

10k 5%



Q8

2N3904



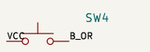
R19

1.5k 5%



D3

TLHR5200



SW4

B3F-4050



R22

10k 5%



R8

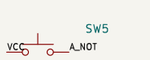
10k 5%



Q6

2N3904

NOT: common-emitter inverter and LED driver



SW5

B3F-4050



R23

10k 5%



R11

10k 5%



Q9

2N3904



R17

2.2k 5%



R12

10k 5%



Q10

2N3904



R20

1.5k 5%



D4

TLHR5200

PET Engenharia Eletrica UFC

Sheet: /

File: 01-logic-gates.kicad_sch

Title: Placa 01 – Portas logicas

Size: A3 | Date: 2026-10-27 | **Rev: A**

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